

Korean Petrochemical Industry

Decarbonization Pathway Analysis

Marginal Abatement Cost Curve (MACC) Model

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Model Version: 2.2

Key Model Parameters

- Baseline Emissions (2025): 52.0 MtCO₂/year
 - Facilities Analyzed: 248
- Technologies: 4 (Heat Pump, RE PPA, NCC-H₂, NCC-Electricity)
 - Time Horizon: 2025-2050
- Maximum Abatement Potential: 49.8 MtCO₂ (95.9%)
 - Scenarios: Conservative, Moderate, Aggressive

Methodology & Key Assumptions

DUAL MACC METHODOLOGY

1. Category A Technologies (Heat Pump, RE PPA)
Traditional MACC Formula:

MACC (\$/tCO₂) = (CAPEX_{annual} + OPEX_{annual} + ΔFuel_{cost}) / Abatement
- CAPEX_{annual}: Capital cost annualized over 20 years at 8% discount rate
 - OPEX_{annual}: Operating costs (3% of CAPEX for Heat Pump, 2% for RE PPA)
 - ΔFuel_{cost}: Net change in fuel costs (often negative = savings)
2. Category B Technologies (NCC-H2, NCC-Electricity)
LCOE Premium Method:

MACC (\$/tCO₂) = (LCOE_{new} - LCOE_{baseline}) / Emission_{intensity}
- LCOE data from: Woo et al. (2025), Green Chemistry
 - Accounts for integrated petrochemical process complexity
 - Includes all capex, opex, and feedstock costs

TECHNOLOGY DETAILS

- Heat Pump (Industrial)
 - Target: All fossil fuel combustion (naphtha, LNG, fuel gas, etc.)
 - COP (Coefficient of Performance): 4.0
 - CAPEX: \$800/kW thermal installed
 - Applicability: 60-100% by product group
 - Result: Highly cost-negative due to fuel savings
- RE PPA (Renewable Energy Power Purchase Agreement)
 - Target: Grid electricity emissions (Scope 2)
 - Mechanism: Switch from grid to dedicated renewable power
 - Declining abatement: Grid decarbonizes 0.40→0.02 tCO₂/MWh (2025→2050)
 - Price trajectory: Premium in 2025 (\$150/MWh), parity 2030+
- NCC-H2 (Hydrogen-Based Naphtha Cracker)
 - Target: Naphtha cracker facilities only (Ethylene, Propylene)
 - H₂ consumption: 0.559 kg H₂ per tCO₂ abated
 - H₂ price: 3.50/kg(2025)→1.50/kg (2050, green H₂)
 - LCOE: \$759/ton ethylene vs \$746/ton baseline (+\$13/ton)
 - Technology maturity: TRL 6-7 (BASF pilot)
- NCC-Electricity (Electric Cracker)
 - Target: Same facilities as NCC-H2 (mutually exclusive)
 - Electricity: 10 MWh/ton ethylene
 - RE power required for zero emissions
 - LCOE: \$737/ton ethylene vs \$746 baseline (-\$9/ton!)
 - Technology maturity: TRL 5-6 (BASF-SABIC development)

KEY PRICE ASSUMPTIONS (2025 → 2050)

- Fossil Fuels (increasing):
 - Naphtha: 15.0/GJ→19.0/GJ (+27%)
 - LNG: 12.0/GJ→17.5/GJ (+46%)
- Electricity (diverging):
 - Grid power: 120/MWh→150/MWh (+25%, infrastructure costs)
 - RE PPA: 150/MWh→110/MWh (-27%, learning curve)
- Hydrogen (declining):
 - Blue H₂: 3.50/kg→2.00/kg (-43%)
 - Green H₂: 5.00/kg→1.50/kg (-70%, 2050 target)
- Grid Emission Factor (decarbonizing):
 - 2025: 0.404 tCO₂/MWh (35% coal, 30% LNG, 25% nuclear, 10% RE)
 - 2030: 0.225 tCO₂/MWh (20% coal, 35% LNG, 30% nuclear, 15% RE)
 - 2050: 0.023 tCO₂/MWh (0% coal, 20% LNG, 10% nuclear, 70% RE)

TECHNOLOGY LEARNING CURVES (CAPEX Reduction)

- All technologies experience cost reductions through deployment and learning:
- Heat Pump: 150M→75M/MtCO₂ (-50%, -2.7%/year)
 - NCC-H₂: 300M→150M/MtCO₂ (-50%, -2.7%/year)
 - NCC-Electricity: 350M→180M/MtCO₂ (-49%, -2.6%/year)
 - RE PPA: 180M→80M/MtCO₂ (-56%, -3.2%/year)

Method: Linear interpolation between milestone years (2025, 2030, 2040, 2050)
Basis: IEA, IRENA, Hydrogen Council projections
Assumption: Time-based learning (conservative vs deployment-based)

OPTIMIZATION CONSTRAINTS

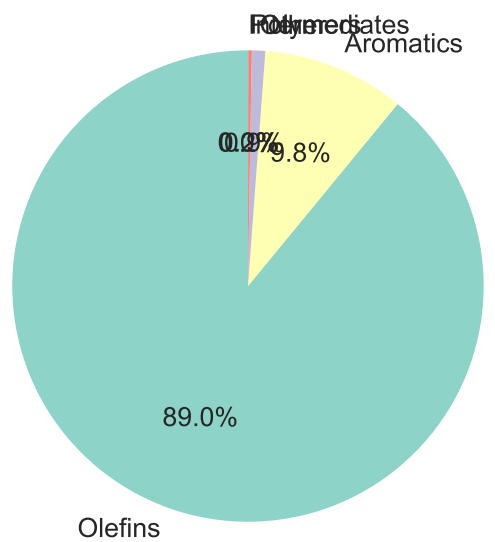
1. Technology Irreversibility
Once deployed, technology cannot be reversed
→ Deployment[t] ≥ Deployment[t-1] for all years
2. Mutual Exclusivity
NCC-H2 and NCC-Electricity target same facilities
→ Can only deploy one per cracker
3. Annual Emission Targets
Linear interpolation between milestone years
→ Conservative: 52 Mt (2025) → 20 Mt (2050), 62% reduction
→ Moderate: 52 Mt (2025) → 10 Mt (2050), 81% reduction
→ Aggressive: 52 Mt (2025) → 5 Mt (2050), 90% reduction
4. Cost Minimization
Deploy least-cost technologies first
Subject to abatement potential limits

DATA SOURCES

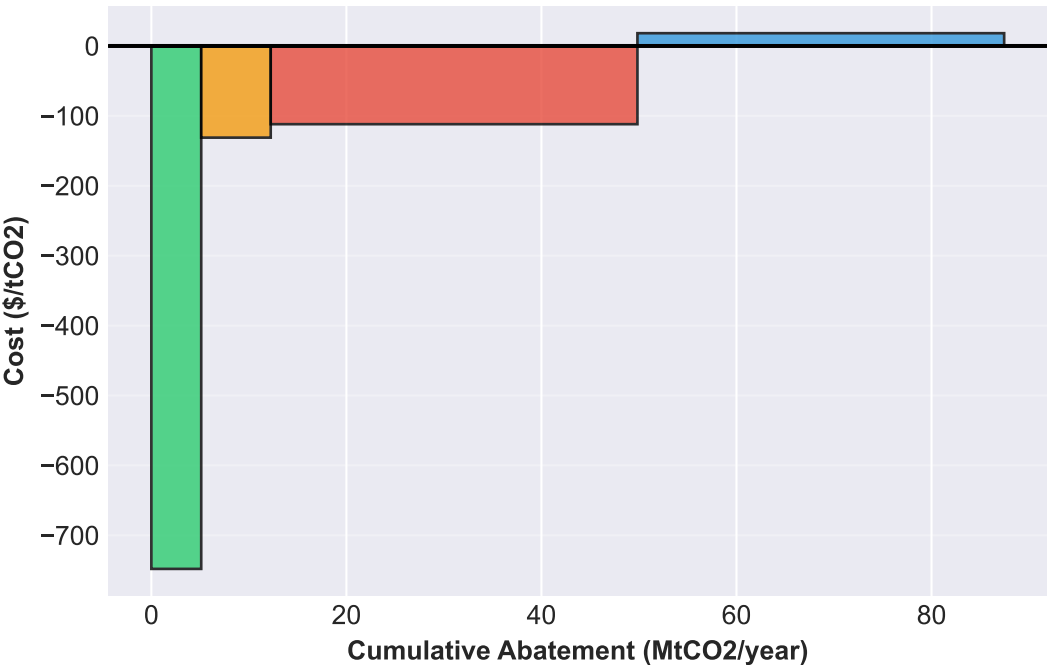
- Baseline emissions: GHG Inventory Center (2021) + growth projections
 - LCOE data: Woo et al. (2025), Green Chemistry, DOI:10.1039/D4GC04538F
 - Heat pump costs: IEA Industrial Heat Pumps (2023)
 - Hydrogen prices: IEA Hydrogen Strategy (2021), Hydrogen Council (2021)
 - Grid factors: Korea 10th Power Supply Plan (2023)
 - RE prices: IRENA Renewable Power Generation Costs (2023)

Executive Summary

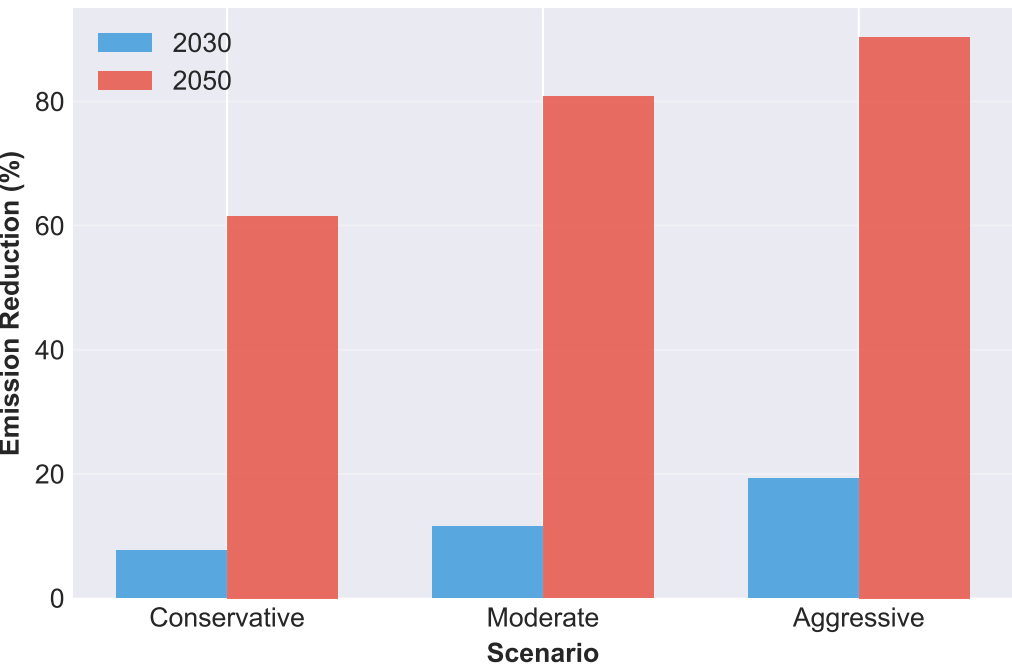
2025 Baseline by Product Group
(52.0 MtCO₂ total)



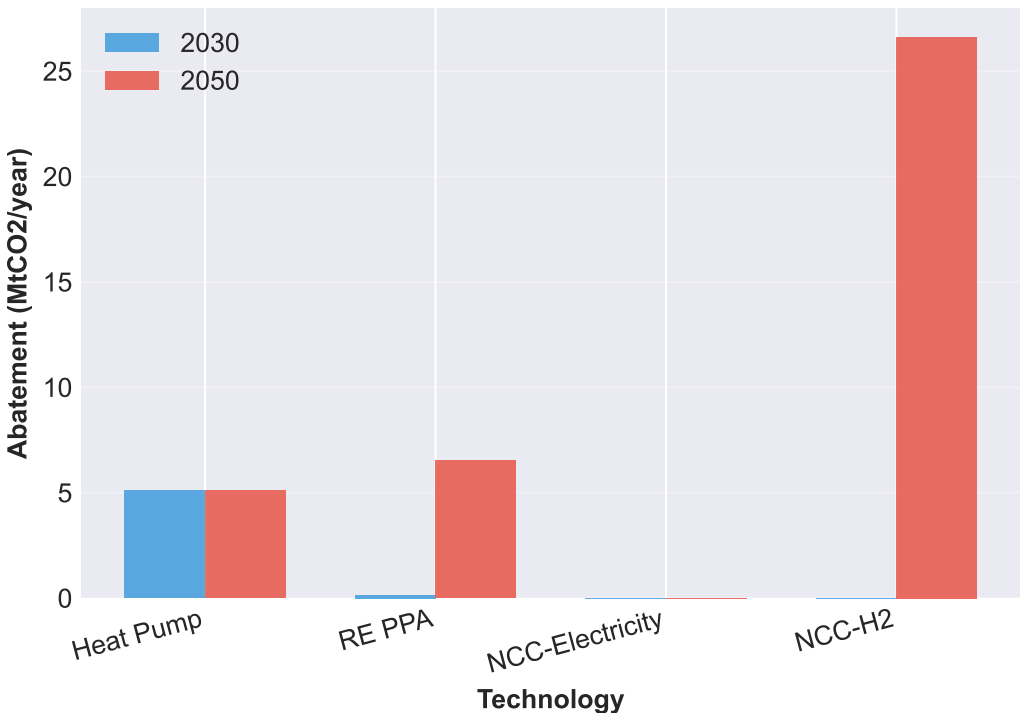
MACC Curve - 2030
(3 cost-negative technologies)



Emission Reduction by Scenario

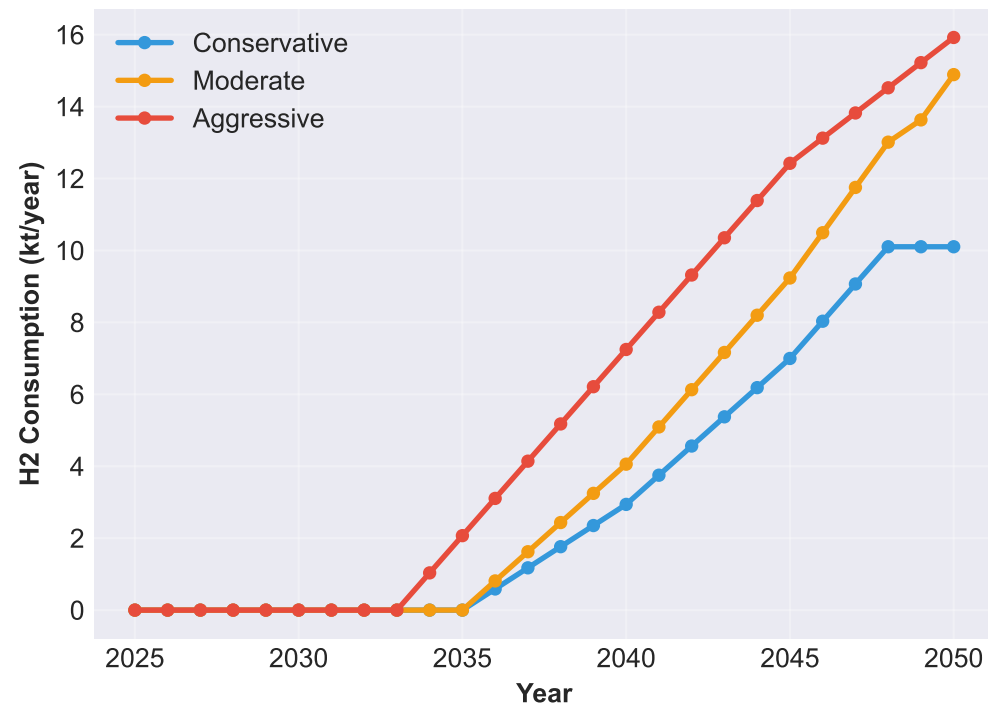


Technology Deployment - Moderate Scenario

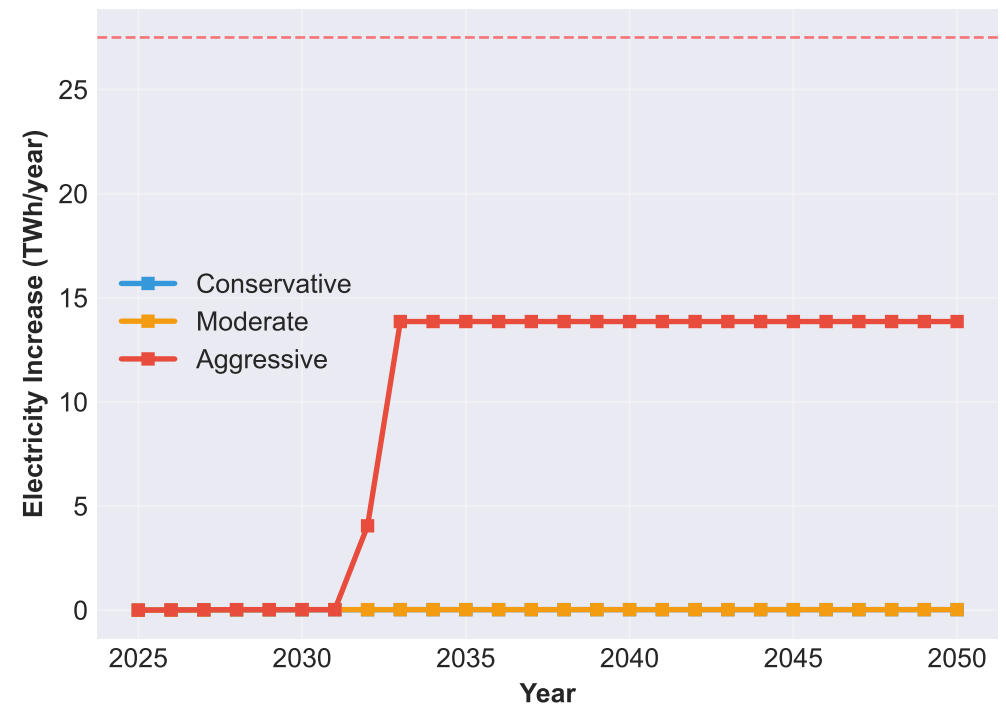


Energy Demand & Investment Analysis

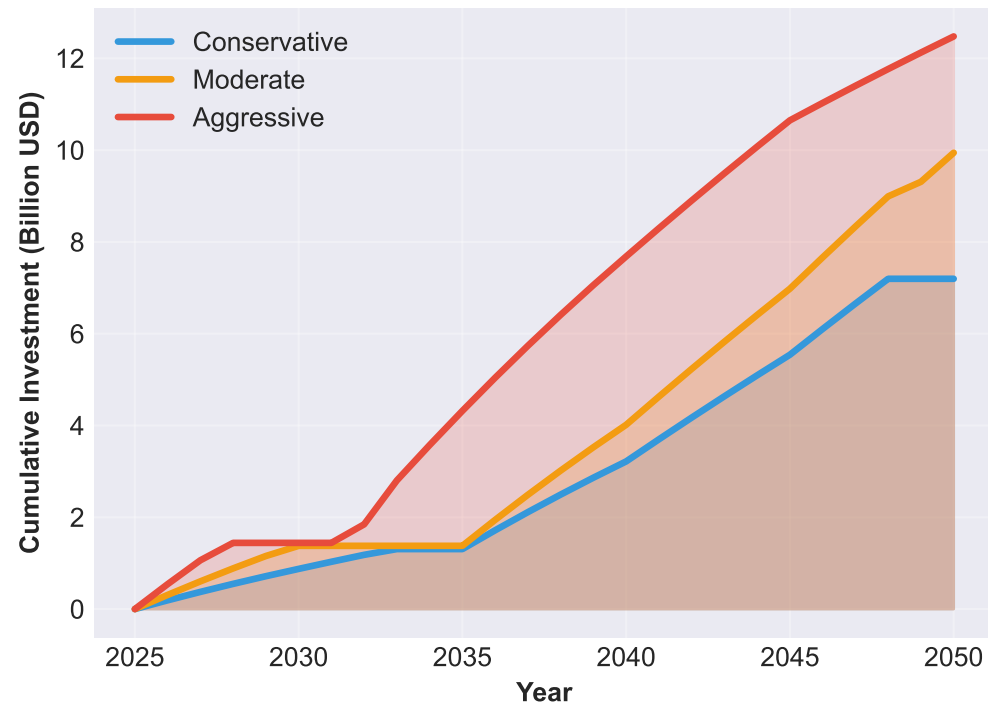
Hydrogen Demand Trajectory



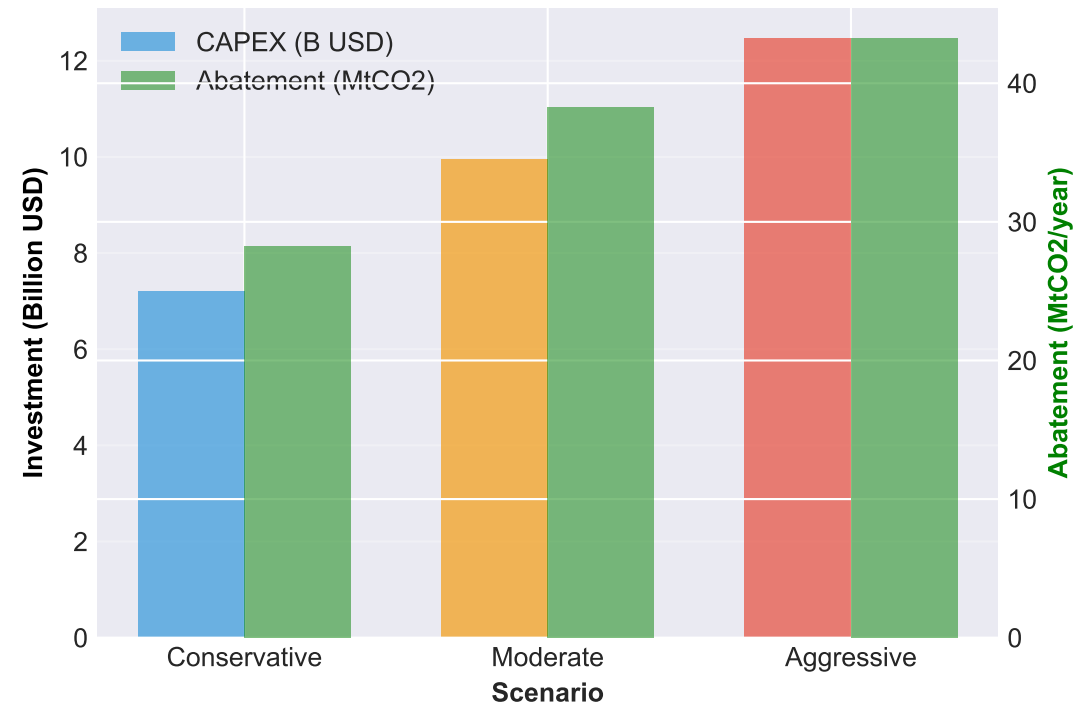
Additional Electricity Demand



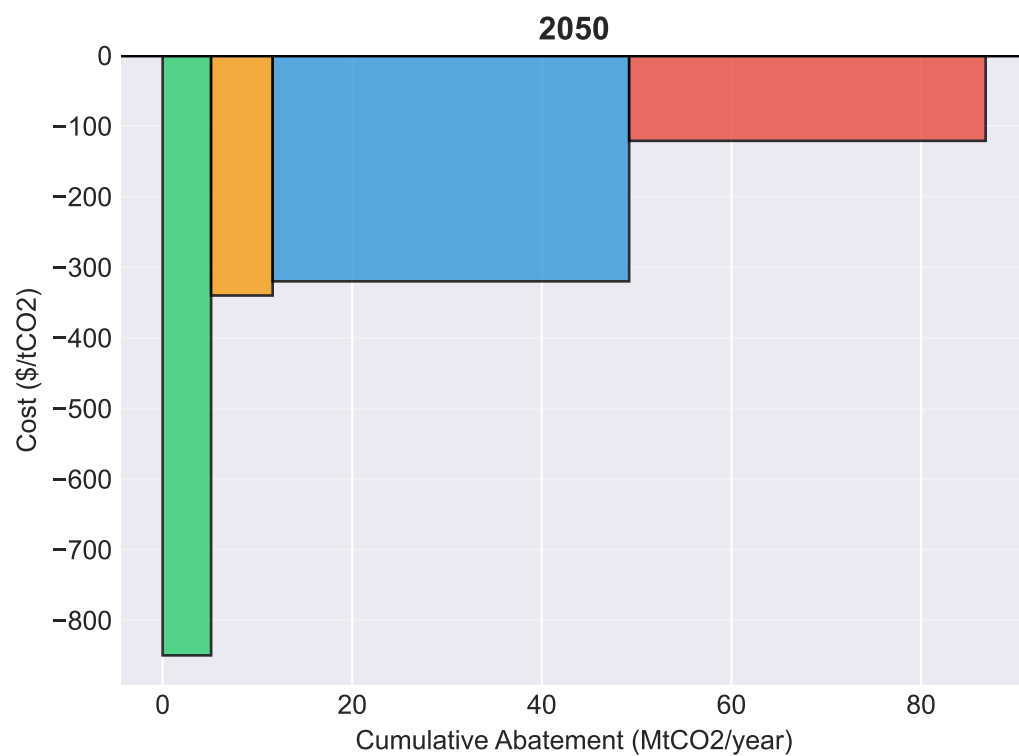
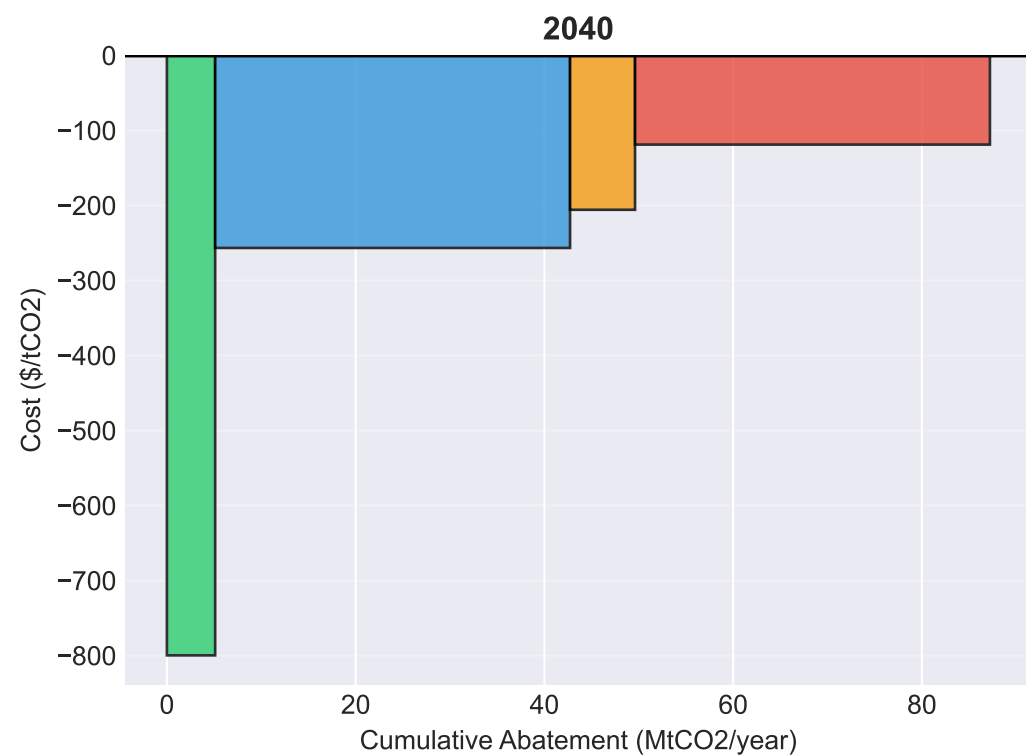
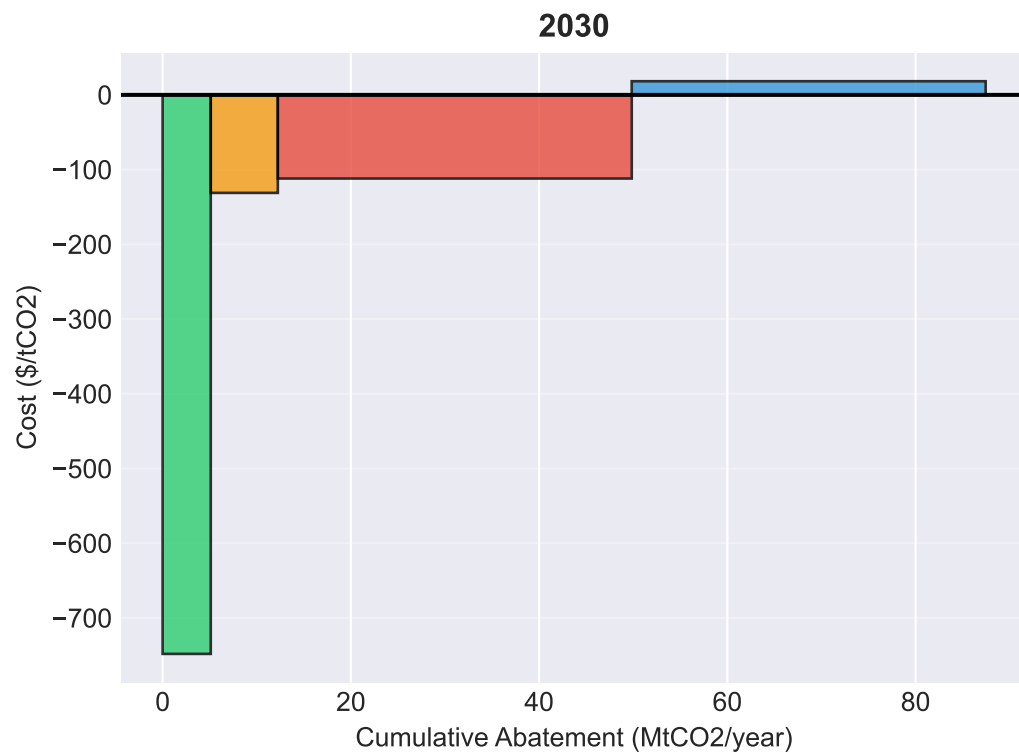
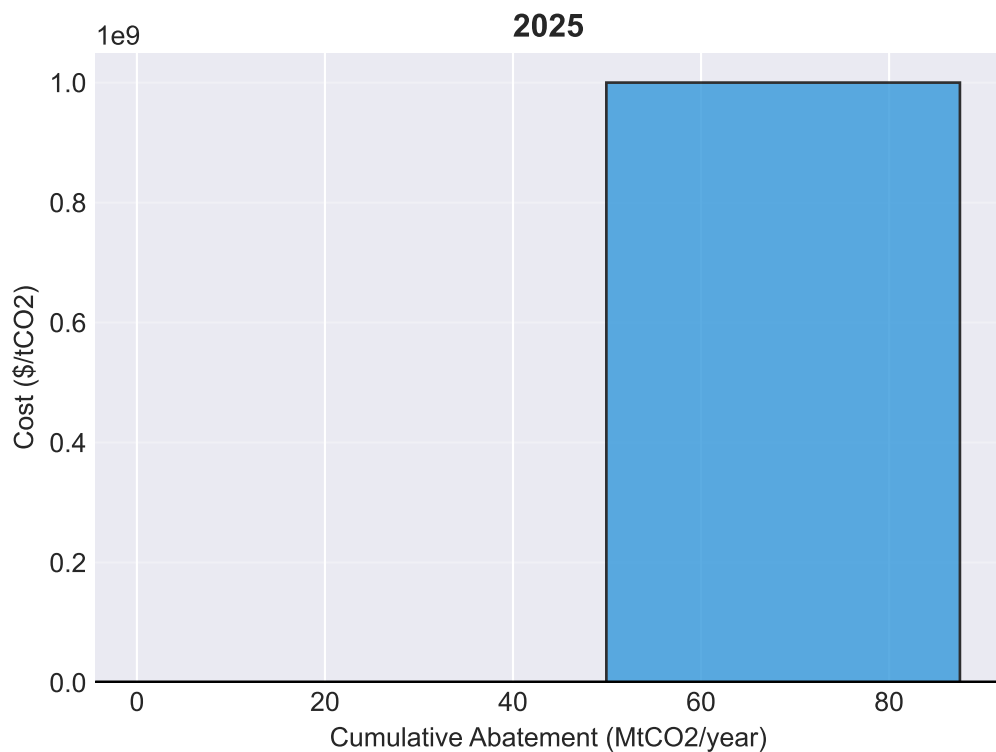
Total CAPEX Requirements



Investment vs Abatement (2050)

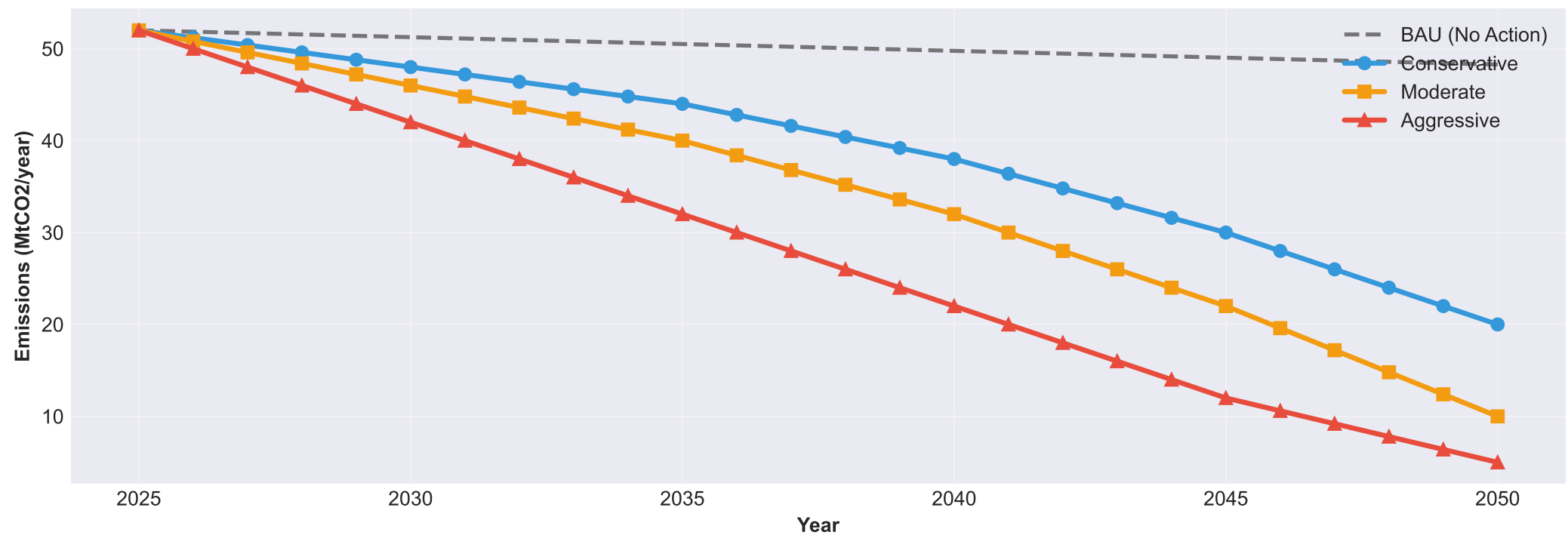


MACC Curve Evolution (2025-2050)

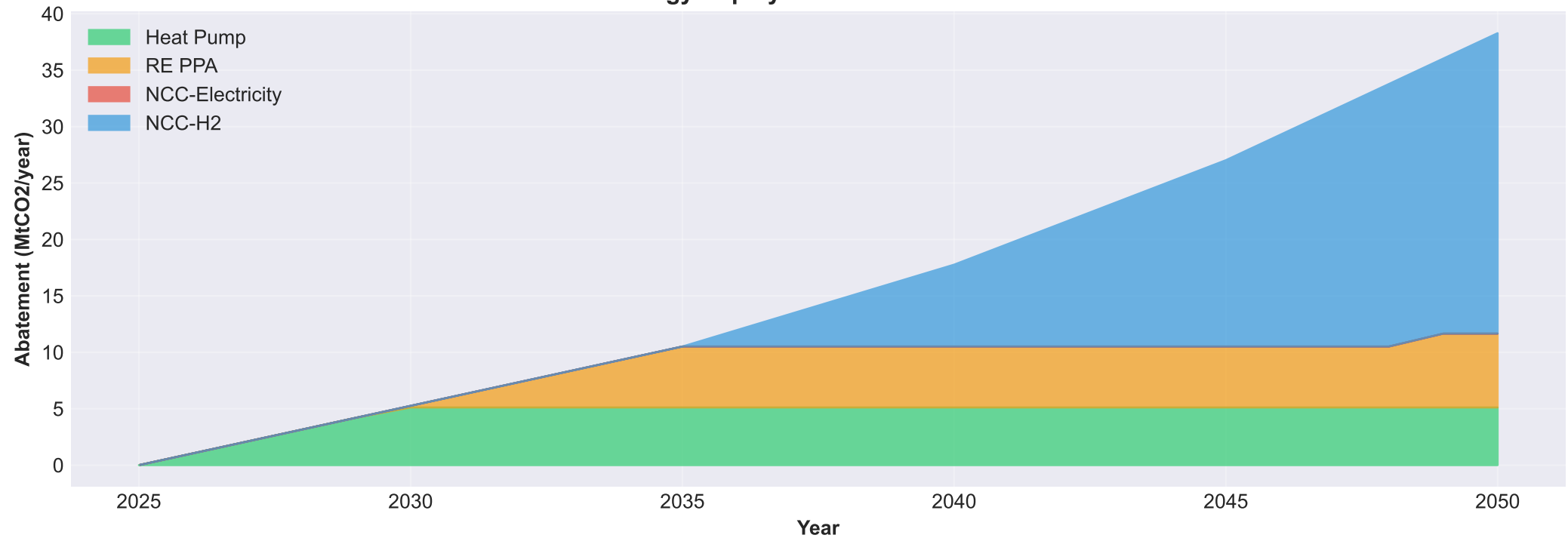


Emission Trajectories by Scenario

Annual Emissions



Technology Deployment - Moderate Scenario



Summary Tables

Scenario Comparison

Scenario	2030 Emis (Mt)	2030 Reduct (%)	2050 Emis (Mt)	2050 Reduct (%)
Conservative	48.0	7.7%	20.0	61.5%
Moderate	46.0	11.5%	10.0	80.8%
Aggressive	42.0	19.2%	5.0	90.4%

Technology Costs - 2030

Technology	Abatement (MtCO2)	CAPEX+OPEX (\$/tCO2)	Fuel Diff (\$/tCO2)	Total Cost (\$/tCO2)
Heat_Pump	5.11	\$12.6	\$-760.6	\$-748.0
NCC-H2	37.60	\$24.0	\$0.0	\$18.3
NCC-Electricity	37.60	\$28.7	\$0.0	\$-111.9
RE_PPA	7.13	\$0.0	\$-131.1	\$-131.1